

Práctica Tema 5

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1 Apartado 2

Implementación del modelo base con el algoritmo MVA en Python3:

```
import math

class Service:
    def __init__(self, name, service, visit):
        self.name, self.S, self.V = name, service, visit
        self.D = self.V * self.S
        self.N, self.R, self.X, self.U = ([0.0] for _ in range(4))

cpu, disc = Service("cpu", 0.03, 8.0), Service("disc", 0.1, 7.0)
devs = [cpu, disc]

N, Z = 10, 8.0
V = sum(dev.V for dev in devs)
D = sum(dev.D for dev in devs)
Db = max(dev.D for dev in devs)
Ns = math.ceil((D + Z) / Db)

sep = "\t"
cols = ("D", "D1", "D2", "Db", "N*")
print(sep.join(cols))
data = (D, cpu.D, disc.D, Db, Ns)
print(sep.join(f"{x:.4f}" for x in data), end="\n\n")

X, R, RT, NW, NZ = ([0.0] for _ in range(5))

cols = ("N", "R1", "R2", "R", "X0", "N1", "N2", "Rt", "Nw", "Rz")
print(sep.join(cols))

for n in range(1, N+1):
    R.append(0.0)
    for dev in devs:
        dev.R.append(dev.S * (1 + dev.N[n-1]))
        R[n] += dev.V * dev.R[n]
    RT.append(R[n] + Z)
    X.append(n / (Z + R[n]))
    NW.append(X[n] * R[n])
    NZ.append(X[n] * Z)
    for dev in devs:
        dev.X.append(X[n] * dev.V)
        dev.N.append(dev.X[n] * dev.R[n])
        dev.U.append(dev.X[n] * dev.S)
```

```
Rt = R[n] + Z
Nw = X[n] * R[n]
Nz = X[n] * Z
data = (cpu.R[n], disc.R[n], R[n], X[n], cpu.N[n], disc.N[n], Rt, Nw, Nz)
print(f"{n}{sep}{sep.join(f'{x:.4f}' for x in data)}")
```

Resultado de su ejecución:

D	D1	D2	Db	N*
0.9400	0.2400	0.7000	0.7000	13.0000

N	R1	R2	R	X0	N1	N2	Rt	Nw	Rz
1	0.0300	0.1000	0.9400	0.1119	0.0268	0.0783	8.9400	0.1051	0.8949
2	0.0308	0.1078	1.0013	0.2222	0.0548	0.1677	9.0013	0.2225	1.7775
3	0.0316	0.1168	1.0705	0.3307	0.0837	0.2703	9.0705	0.3541	2.6459
4	0.0325	0.1270	1.1493	0.4372	0.1137	0.3888	9.1493	0.5025	3.4975
5	0.0334	0.1389	1.2394	0.5412	0.1446	0.5261	9.2394	0.6707	4.3293
6	0.0343	0.1526	1.3430	0.6422	0.1764	0.6860	9.3430	0.8624	5.1376
7	0.0353	0.1686	1.4626	0.7398	0.2089	0.8731	9.4626	1.0819	5.9181
8	0.0363	0.1873	1.6013	0.8332	0.2417	1.0925	9.6013	1.3342	6.6658
9	0.0373	0.2092	1.7628	0.9219	0.2747	1.3503	9.7628	1.6250	7.3750
10	0.0382	0.2350	1.9511	1.0049	0.3074	1.6533	9.9511	1.9607	8.0393